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What is digestion?

[2 marks]

One species of fungus digests cellulose using two types of enzyme, endocellulases and exocellulases.

Endocellulases act in the middle of the cellulose molecule and exocellulases act at the ends of the cellulose molecule.

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Endocellulases and exocellulases act at different places on cellulose molecules.

Suggest why.

[2 marks]

Turn over ►

A scientist prepared the following mixtures:

- 15 g cellulose with 0.2 mol dm^{-3} endocellulase
- 15 g cellulose with 0.2 mol dm^{-3} exocellulase
- 15 g cellulose with 0.2 mol dm^{-3} endocellulase and 0.2 mol dm^{-3} exocellulase.

The mixtures had identical total volumes. She determined the mass of cellulose remaining after 48 hours.

Her results are shown in **Table 4**.

Table 4

Time / hours	Mass of cellulose remaining / g		
	Endocellulase	Exocellulase	Endocellulase + exocellulase
48	11.9	14.8	9.2

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Use information from **Table 4** to calculate the rate of digestion of cellulose when both enzymes are present.

Give your answer in g min^{-1} and in standard form.
Show your working.

[2 marks]

Answer = _____ g min^{-1}



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The scientist used the same concentration of endocellulase and exocellulase in the mixtures. The rate of digestion of cellulose is greatest when both enzymes are present.

Suggest why.

[2 marks]

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The scientist could have expressed her results as the percentage loss in mass of cellulose.

In the space, write the equation for calculating the percentage loss in mass.

[1 mark]

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Turn over for the next question

Turn over ►

